

Final MOSC Project - Insurance Dilemma - Caleb Ramsey

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Introduction

This project studies the five-year risk profile of a hypothetical insurance company using simulation. The company begins with \$1,000,000 in assets, collects annual premiums from 1,000 customers, and faces random claim events each year. Claim sizes are modeled with a Pareto distribution, which is useful in insurance settings because it allows for rare but very large losses.

The main goals of this analysis are to:

1. Generate Pareto-distributed claim sizes using the inverse CDF method.
2. Simulate the company's assets over a five-year period.
3. Estimate the probability of bankruptcy and expected assets after five years.
4. Compare the original asset process with a profit-taking version where excess assets are paid out to shareholders at the end of each year.

Libraries

```
library(tidyverse)
library(scales)
```

Problem Setup

```
alpha <- 3
beta  <- 100000

Z0 <- 1000000

n_customer <- 1000
p_claim    <- 0.10

annual_premium <- 5500
n_years        <- 5
n_sim          <- 1000
```

The expected value of a single claim is (Pareto Type II expected value formula):

$$E(X) = \frac{\beta}{\alpha - 1}$$

For this analysis, the expected claim size is:

```
expected_claim_size <- beta / (alpha - 1)
expected_claim_size
```

```
## [1] 50000
```

Before running the full simulation, this gives a useful baseline for understanding the company's annual risk exposure.

```
expected_number_claims <- n_customer * p_claim
expected_total_claims <- expected_number_claims * expected_claim_size
expected_premium_income <- n_customer * annual_premium
expected_net_annual_change <- expected_premium_income - expected_total_claims

expected_summary <- tibble(
  Metric = c(
    "Expected # of claims per year",
    "Expected total annual claims",
    "Annual premium income",
    "Expected net annual change"
  ),
  Value = c(
    expected_number_claims,
    expected_total_claims,
    expected_premium_income,
    expected_net_annual_change
  )
)

expected_summary
```

```
## # A tibble: 4 x 2
##   Metric                                Value
##   <chr>                                <dbl>
## 1 Expected # of claims per year         100
## 2 Expected total annual claims 5000000
## 3 Annual premium income           5500000
## 4 Expected net annual change         500000
```

Before simulation, the company appears profitable on average. With 1,000 customers, roughly 100 claims are expected each year, leading to about \$5,000,000 in expected annual claims. Since annual premium income is \$5,500,000, the expected net gain is \$500,000 per year. However, average profitability does not guarantee survival as claim sizes are modeled with a heavy-tailed Pareto distribution.

Part 1: Pareto Random Variable Generation

The claim size random variable follows a Pareto distribution with CDF:

$$F_X(x) = 1 - \frac{\beta^\alpha}{(x + \beta)^\alpha}, \quad x \geq 0$$

To simulate claim sizes, the inverse CDF method is employed. The idea is to generate a uniform random variable $U \sim \text{Uniform}(0,1)$, set $U = F_X(x)$, and solve for x .

Pareto Generator

```
rPareto <- function(n_sample, alpha, beta) {

  # Generate Uniform(0,1) random variables
  u <- runif(n_sample)

  # Inverse CDF derivation:
  #
  # U = 1 - beta^alpha / (x + beta)^alpha
  #
  # Rearranging:
  # 1 - U = beta^alpha / (x + beta)^alpha
  #
  # Solving for x:
  # x = beta * ((1 - U)^(-1/alpha) - 1)

  x <- beta * ((1 - u)^(-1 / alpha) - 1)

  return(x)
}
```

True Density Function

```
dPareto <- function(x, alpha, beta) {
  ifelse(
    x >= 0,
    (alpha * beta^alpha) / ((x + beta)^(alpha + 1)),
    0
  )
}
```

Simulated Claim Sizes

```
set.seed(123)

pareto_sample <- rPareto(
  n_sample = 1000,
  alpha = alpha,
  beta = beta
)
```

```
claim_summary <- tibble(
  Statistic = c("Mean", "Median", "Standard Deviation", "Minimum", "Maximum"),
  Value = c(
    mean(pareto_sample),
    median(pareto_sample),
    sd(pareto_sample),
    min(pareto_sample),
    max(pareto_sample)
  )
)

claim_summary
```

```
## # A tibble: 5 x 2
##   Statistic      Value
##   <chr>         <dbl>
## 1 Mean          48454.
## 2 Median         25159.
## 3 Standard Deviation 72889.
## 4 Minimum         15.5
## 5 Maximum       1088629.
```

Table 1. Summary statistics for 1,000 simulated Pareto claim sizes.

```
ggplot(tibble(claim_size = pareto_sample), aes(x = claim_size)) +
  geom_histogram(aes(y = after_stat(density)), bins = 40, boundary = 0) +
  stat_function(
    fun = function(x) dPareto(x, alpha = alpha, beta = beta),
    linewidth = 1
  ) +
  scale_x_continuous(labels = dollar_format()) +
  labs(
    title = "Simulated Pareto Claim Sizes",
    subtitle = "Histogram of simulated claims with the true density curve overlaid",
    x = "Claim Size",
    y = "Density"
  ) +
  theme_minimal(base_size = 12)
```

The distribution of simulated claim sizes is heavily right-skewed, with most claims concentrated at relatively small values near the left side of the histogram. The density decreases rapidly as claim size increases, but the distribution also contains a long right tail extending toward extremely large claims, including some near or above \$1 million.

In the context of insurance, this long right tail represents rare but catastrophic claim events. Most customers generate manageable claim amounts, allowing the company to remain profitable in many years. However, a small number of unusually large claims can create severe financial losses that dramatically reduce company assets or even lead to bankruptcy. This heavy-tailed behavior is one of the main reasons the company can still face substantial insolvency risk despite having positive expected profits overall.

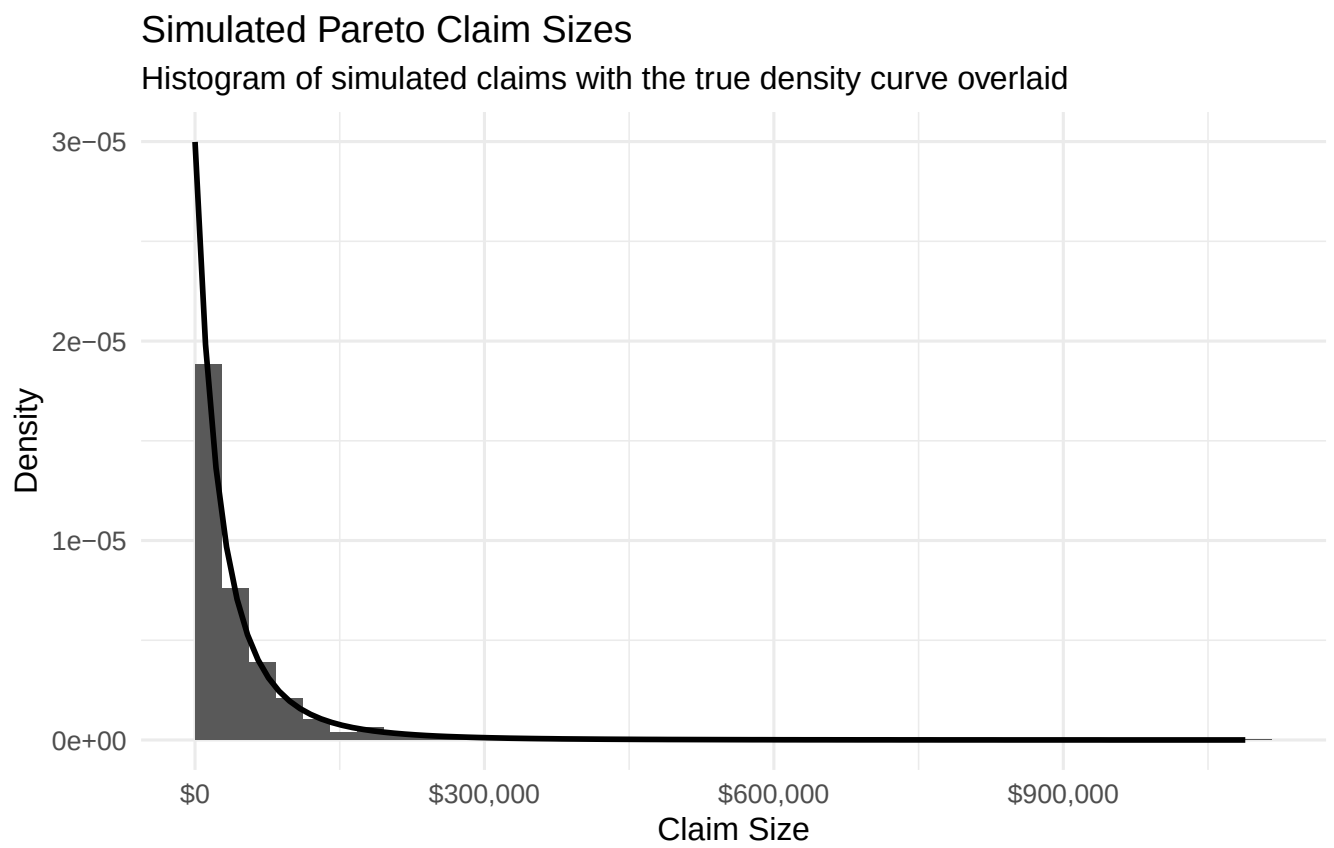


Figure 1: Histogram of simulated Pareto claim sizes with the true density curve overlaid.

Part 2: Simulating Company Assets

The company begins with \$1,000,000 in assets. Each year, customers pay premiums at the beginning of the year. Then some customers make claims, and the company pays those claim amounts. If assets reach zero, the company is considered bankrupt and remains at zero.

Asset Simulation Function

```
assets <- function(t, alpha, beta, Z0, n_customer, p, annual_premium) {  
  
  Z <- numeric(t + 1)  
  Z[1] <- Z0  
  
  for (year in 1:t) {  
  
    if (Z[year] <= 0) {  
      Z[year + 1] <- 0  
  
    } else {  
  
      premium_income <- n_customer * annual_premium  
  
      claim_indicators <- rbinom(  
        n = n_customer,  
        size = 1,  
        prob = p  
      )  
  
      n_claims <- sum(claim_indicators)  
  
      if (n_claims > 0) {  
        claim_sizes <- rPareto(  
          n_sample = n_claims,  
          alpha = alpha,  
          beta = beta  
        )  
  
        total_claims <- sum(claim_sizes)  
  
      } else {  
        total_claims <- 0  
      }  
  
      Z[year + 1] <- max(  
        Z[year] + premium_income - total_claims,  
        0  
      )  
    }  
  }  
  
  return(Z)  
}
```

One Five-Year Asset Trajectory

```
set.seed(123)

one_path <- assets(
  t = n_years,
  alpha = alpha,
  beta = beta,
  Z0 = Z0,
  n_customer = n_customer,
  p = p_claim,
  annual_premium = annual_premium
)

one_path_df <- tibble(
  Year = 0:n_years,
  Assets = one_path
)

one_path_df
```

```
## # A tibble: 6 x 2
##   Year  Assets
##   <int>   <dbl>
## 1     0 1000000
## 2     1 2990856.
## 3     2 3482599.
## 4     3 2800375.
## 5     4 1508966.
## 6     5 3306647.
```

Table 2. Simulated company assets over one five-year trajectory.

```
ggplot(one_path_df, aes(x = Year, y = Assets, color = Assets)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2.5) +
  scale_color_gradient2(
    low = "#b22222",
    mid = "gray45",
    high = "#1b7837",
    midpoint = Z0
  ) +
  scale_y_continuous(labels = dollar_format()) +
  scale_x_continuous(breaks = 0:n_years) +
  labs(
    title = "One Simulated Asset Trajectory",
    subtitle = "Company assets from year 0 through year 5",
    x = "Year",
    y = "Assets",
    color = "Assets"
  ) +
  theme_minimal(base_size = 12)
```

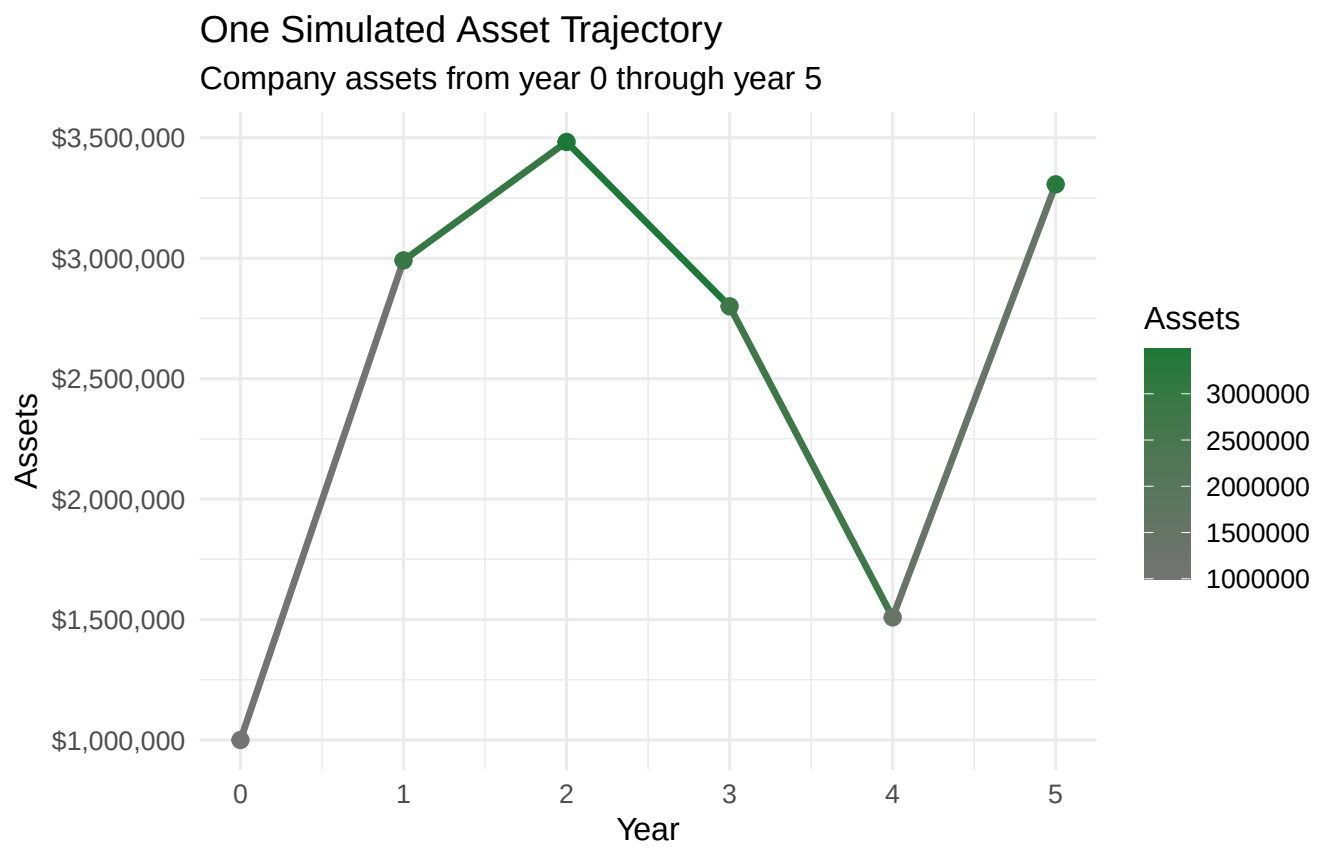


Figure 2: One simulated five-year asset trajectory for the insurance company.

In this single trajectory, the company experienced strong early growth, with assets increasing to roughly \$3.5 million by years 1 and 2. However, years 3 and 4 showed a noticeable decline, likely caused by several unusually large insurance claims. Despite those losses, the company had accumulated enough reserves during the profitable early years to avoid bankruptcy and eventually recovered, ending near \$5 million in assets by year 5.

However, one trajectory alone is not enough to estimate the company's true level of risk. A single simulation only represents one possible outcome from a highly random process. If the large claims in this trajectory had occurred earlier, before the company built up reserves, the company may have gone bankrupt instead. Because claim sizes follow a heavy-tailed Pareto distribution, rare but extreme losses can dramatically change outcomes. Running many simulations allows us to estimate the overall probability of bankruptcy and better understand the full range of possible financial outcomes.

Generate 1,000 Asset Trajectories

```
set.seed(123)

all_paths <- matrix(
  NA_real_,
  nrow = n_sim,
  ncol = n_years + 1
)

for (i in 1:n_sim) {
  all_paths[i, ] <- assets(
    t = n_years,
    alpha = alpha,
    beta = beta,
    Z0 = Z0,
    n_customer = n_customer,
    p = p_claim,
    annual_premium = annual_premium
  )
}

all_paths_df <- as_tibble(all_paths)
names(all_paths_df) <- paste0("Year_", 0:n_years)

all_paths_long <- all_paths_df %>%
  mutate(Simulation = row_number()) %>%
  pivot_longer(
    cols = starts_with("Year_"),
    names_to = "Year",
    values_to = "Assets"
  ) %>%
  mutate(
    Year = parse_number(Year),
    Bankrupt = Assets == 0
  )

ggplot(all_paths_long, aes(x = Year, y = Assets, group = Simulation)) +
  geom_line(alpha = 0.12) +
```

```

scale_y_continuous(labels = dollar_format()) +
scale_x_continuous(breaks = 0:n_years) +
labs(
  title = "1,000 Simulated Asset Trajectories",
  subtitle = "Each line represents one possible five-year path for the company",
  x = "Year",
  y = "Assets"
) +
theme_minimal(base_size = 12)

```

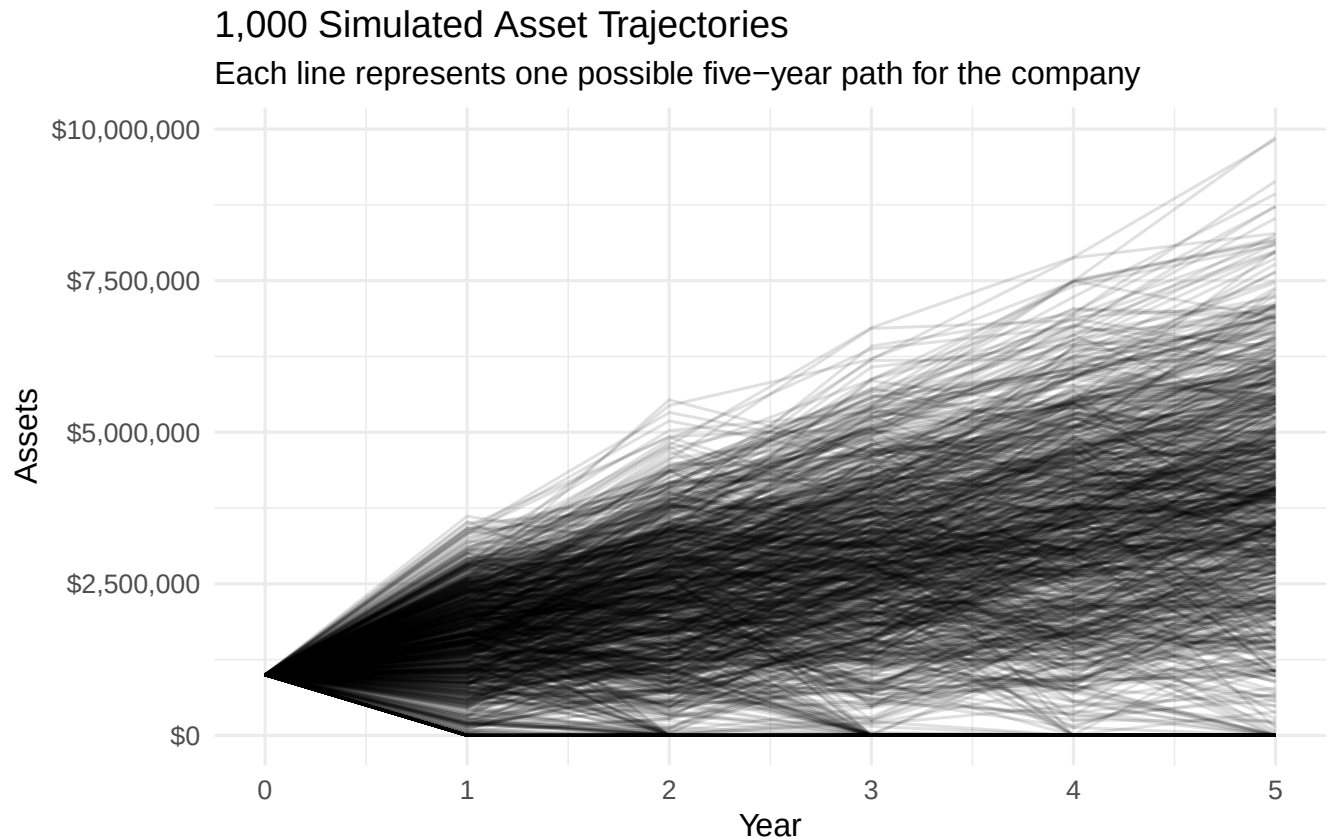


Figure 3: One thousand simulated asset trajectories over five years.

Estimated Bankruptcy Probability and Expected Final Assets

```

ruin_indicator <- apply(
  all_paths,
  1,
  function(x) any(x == 0)
)

prob_ruin <- mean(ruin_indicator)

```

```

final_assets <- all_paths[, n_years + 1]
expected_assets <- mean(final_assets)

main_results <- tibble(
  Metric = c(
    "Estimated probability of bankruptcy by year 5",
    "Expected assets at end of year 5"
  ),
  Estimate = c(prob_ruin, expected_assets)
)

main_results

```

```

## # A tibble: 2 x 2
##   Metric                                Estimate
##   <chr>                                <dbl>
## 1 Estimated probability of bankruptcy by year 5      0.163
## 2 Expected assets at end of year 5                3394184.

```

Table 3. Estimated bankruptcy probability and expected final assets from 1,000 simulated trajectories.

```

ggplot(
  tibble(Final_Assets = final_assets),
  aes(
    x = Final_Assets,
    fill = after_stat(x)
  )
) +
  geom_histogram(
    bins = 40,
    color = "white",
    linewidth = 0.2
  ) +
  scale_fill_gradient2(
    low = "#b22222",
    mid = "#d4a72c",
    high = "#1b7837",
    midpoint = 2500000,
    guide = "none"
  ) +
  geom_vline(
    xintercept = 0,
    color = "firebrick",
    linewidth = 1
  ) +
  scale_x_continuous(labels = dollar_format()) +
  labs(
    title = "Distribution of Final Assets",
    subtitle = "Company assets at the end of year 5 across 1,000 simulations",
    x = "Final Assets",
    y = "Number of Simulations"
  )

```

```
) +
theme_minimal(base_size = 12)
```

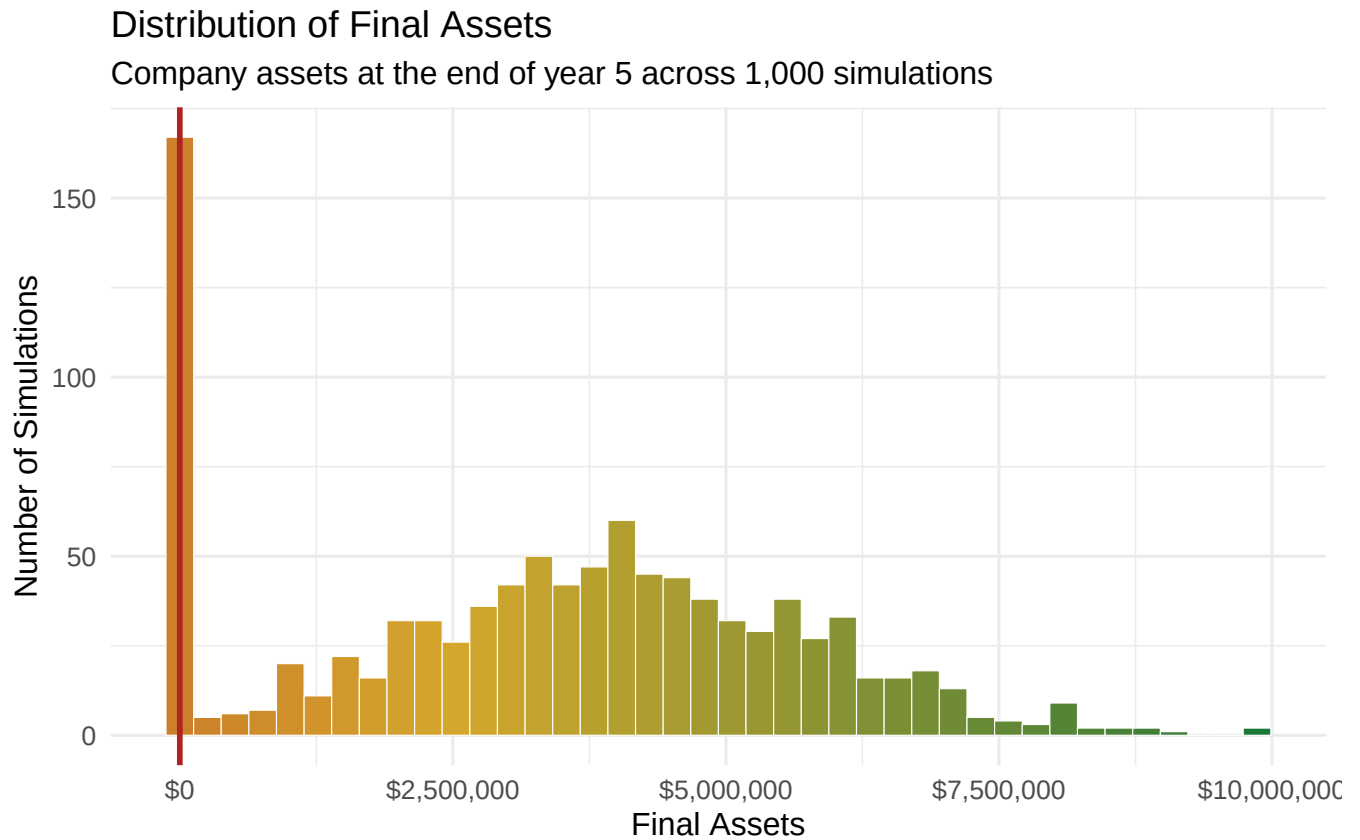


Figure 4: Distribution of company assets at the end of year 5 across 1,000 simulations.

Across the 1,000 simulations, the estimated probability that the company goes bankrupt within five years was approximately 16.3%. Although the expected assets at the end of year 5 were about \$3.39 million, the simulations show substantial variability in outcomes. The histogram of final assets demonstrates that while many simulations ended with several million dollars in assets, bankruptcy remained the single most common individual outcome.

The results are heavily influenced by the Pareto-distributed claim sizes. Most claims are moderate, but the heavy-tailed distribution allows for occasional extremely large claims that can dramatically reduce company assets in a short period of time. This is especially dangerous in the early years of the simulation, when the company has had less time to accumulate reserves through premium income. A few catastrophic claims can therefore outweigh many profitable years, which explains why the company can have positive expected growth overall while still facing a meaningful probability of insolvency.

The simulated trajectories also highlight this volatility. Many paths show relatively stable growth over time, while others collapse rapidly following large claim events. This demonstrates how insurance companies are exposed not only to average claim behavior, but also to rare, high-severity losses.

Part 3: Profit-Taking Policy

Now suppose the company takes profits at the end of each year. If assets exceed \$1,000,000, the excess is paid out to shareholders. If assets are less than or equal to \$1,000,000, shareholders receive nothing that year.

This changes the risk profile because the company is no longer allowed to build a large surplus above the original asset level.

Profit-Taking Asset Function

```
assets_profit <- function(t, alpha, beta, Z0, n_customer, p, annual_premium) {  
  
  Z <- numeric(t + 1)  
  Z[1] <- Z0  
  
  for (year in 1:t) {  
  
    if (Z[year] <= 0) {  
      Z[year + 1] <- 0  
  
    } else {  
  
      premium_income <- n_customer * annual_premium  
  
      claim_indicators <- rbinom(  
        n = n_customer,  
        size = 1,  
        prob = p  
      )  
  
      n_claims <- sum(claim_indicators)  
  
      if (n_claims > 0) {  
        claim_sizes <- rPareto(  
          n_sample = n_claims,  
          alpha = alpha,  
          beta = beta  
        )  
  
        total_claims <- sum(claim_sizes)  
  
      } else {  
        total_claims <- 0  
      }  
  
      # Assets before the profit-taking rule is applied  
      temp_assets <- Z[year] + premium_income - total_claims  
  
      # Bankruptcy comes first  
      if (temp_assets <= 0) {  
        Z[year + 1] <- 0  
      }  
    }  
  }  
}
```

```

    } else {

      # If assets exceed the starting level, pay out the excess
      if (temp_assets > Z0) {
        Z[year + 1] <- Z0
      } else {
        Z[year + 1] <- temp_assets
      }
    }
  }
}

return(Z)
}

```

Simulate 1,000 Profit-Taking Trajectories

```

set.seed(123)

all_paths_profit <- matrix(
  NA_real_,
  nrow = n_sim,
  ncol = n_years + 1
)

for (i in 1:n_sim) {
  all_paths_profit[i, ] <- assets_profit(
    t = n_years,
    alpha = alpha,
    beta = beta,
    Z0 = Z0,
    n_customer = n_customer,
    p = p_claim,
    annual_premium = annual_premium
  )
}

all_paths_profit_df <- as_tibble(all_paths_profit)
names(all_paths_profit_df) <- paste0("Year_", 0:n_years)

all_paths_profit_long <- all_paths_profit_df %>%
  mutate(Simulation = row_number()) %>%
  pivot_longer(
    cols = starts_with("Year_"),
    names_to = "Year",
    values_to = "Assets"
  ) %>%
  mutate(Year = parse_number(Year))

ggplot(all_paths_profit_long, aes(x = Year, y = Assets, group = Simulation)) +
  geom_line(alpha = 0.12) +

```

```

scale_y_continuous(labels = dollar_format()) +
scale_x_continuous(breaks = 0:n_years) +
labs(
  title = "1,000 Simulated Asset Trajectories With Profit-Taking",
  subtitle = "Excess assets above $1,000,000 are paid out at the end of each year",
  x = "Year",
  y = "Assets"
) +
theme_minimal(base_size = 12)

```

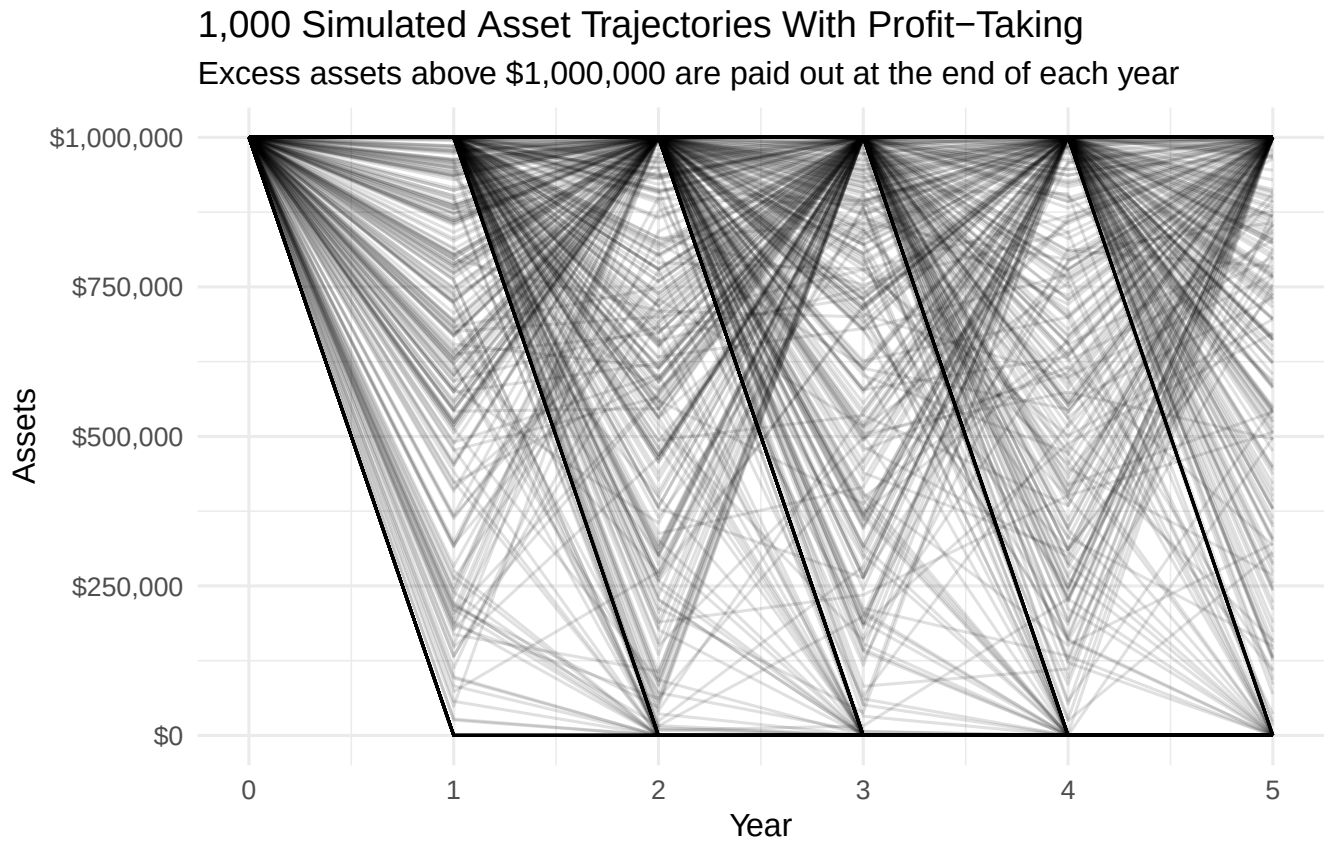


Figure 5: One thousand simulated asset trajectories under the profit-taking policy.

Profit-Taking Results

```

ruin_profit <- apply(
  all_paths_profit,
  1,
  function(x) any(x == 0)
)

prob_ruin_profit <- mean(ruin_profit)

```

```

final_assets_profit <- all_paths_profit[, n_years + 1]
expected_assets_profit <- mean(final_assets_profit)

profit_results <- tibble(
  Metric = c(
    "Estimated probability of bankruptcy by year 5",
    "Expected assets at end of year 5"
  ),
  Estimate = c(prob_ruin_profit, expected_assets_profit)
)

profit_results

```

```

## # A tibble: 2 x 2
##   Metric                                Estimate
##   <chr>                                <dbl>
## 1 Estimated probability of bankruptcy by year 5      0.343
## 2 Expected assets at end of year 5                592788.

```

Table 4. Estimated bankruptcy probability and expected final assets under the profit-taking policy.

Comparison of Policies

```

policy_comparison <- tibble(
  Policy = c("No profit-taking", "Profit-taking"),
  `Probability of Bankruptcy` = c(prob_ruin, prob_ruin_profit),
  `Expected Final Assets` = c(expected_assets, expected_assets_profit)
)

policy_comparison

```

```

## # A tibble: 2 x 3
##   Policy          'Probability of Bankruptcy' 'Expected Final Assets'
##   <chr>                                <dbl>                <dbl>
## 1 No profit-taking                0.163                3394184.
## 2 Profit-taking                  0.343                592788.

```

Table 5. Comparison of estimated bankruptcy risk and expected final assets under both policies.

```

comparison_final_assets <- bind_rows(
  tibble(Policy = "No profit-taking", Final_Assets = final_assets),
  tibble(Policy = "Profit-taking", Final_Assets = final_assets_profit)
)

ggplot(
  comparison_final_assets,
  aes(x = Final_Assets, fill = Policy)
)

```



```

) +
  geom_histogram(
    bins = 40,
    color = "white",
    linewidth = 0.2,
    alpha = 0.9
  ) +
  facet_wrap(~ Policy, scales = "free_y") +
  scale_fill_manual(
    values = c(
      "No profit-taking" = "#2e7d32",
      "Profit-taking" = "#c77c02"
    )
  ) +
  geom_vline(xintercept = 0, color = "firebrick", linewidth = 1) +
  scale_x_continuous(labels = dollar_format()) +
  labs(
    title = "Final Asset Distributions by Policy",
    subtitle = "Comparing asset outcomes after five years",
    x = "Final Assets",
    y = "Number of Simulations"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    legend.position = "none"
  )

```

The two policies produce substantially different outcomes. Under the original policy without profit-taking, the company experiences a much wider range of final asset values, including many high-surplus outcomes. In contrast, the profit-taking policy produces a distribution that is much more concentrated around either bankruptcy or approximately \$1 million in assets, since excess capital is continuously paid out to shareholders rather than retained by the company.

The simulations show that profit-taking substantially increases bankruptcy risk. Under the original policy, the estimated probability of bankruptcy within five years was approximately 16.3%. Under the profit-taking policy, the bankruptcy probability increased to roughly 34.3%, more than doubling the risk of insolvency. The expected final assets also decreased dramatically, falling from roughly \$3.4 million under the original policy to approximately \$600,000 with profit-taking.

Limiting surplus capital matters because retained assets act as a financial buffer against unusually large claims. Since claim sizes follow a heavy-tailed Pareto distribution, a small number of catastrophic claims can severely damage the company's finances. Under the original policy, profitable years allow the company to accumulate reserves that help absorb future losses. Under the profit-taking policy, those reserves are removed each year, leaving the company much more vulnerable to bankruptcy when large claims occur.

Conclusion

In this project, I used Monte Carlo simulation to estimate the risk of bankruptcy for an insurance company with Pareto-distributed claim sizes. The analysis shows how even a company that appears profitable on average can still face meaningful bankruptcy risk when claim sizes are heavy-tailed. The profit-taking policy changes the asset process by preventing the company from retaining excess capital, which affects both expected final assets and the probability of bankruptcy.

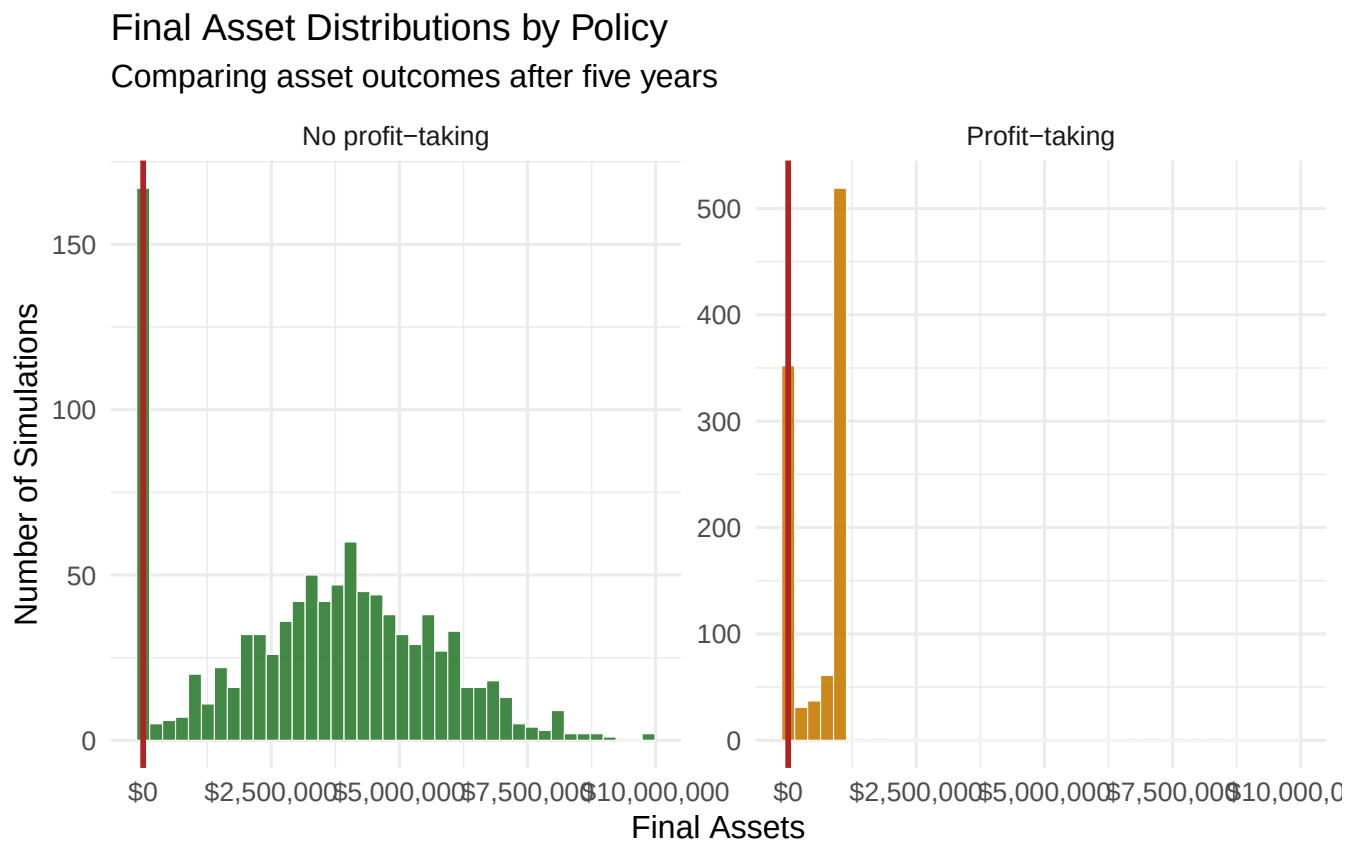


Figure 6: Comparison of final asset distributions under the original and profit-taking policies.

Overall, the no-profit-taking approach produces stronger outcomes over the five-year simulation period. Without profit-taking, the company ended with approximately \$3.4 million in expected final assets and went bankrupt in 16.3% of the 1,000 simulations. In contrast, the profit-taking policy ended in bankruptcy in 34.3% of simulations and produced only about \$590,000 in expected final assets.

The difference comes from how each policy handles surplus capital. Under the profit-taking policy, assets are effectively capped at \$1 million because any excess is paid out to shareholders at the end of each year. This prevents the company from building reserves that could help absorb future large claims. Under the no-profit-taking policy, successful years allow the company to accumulate surplus, creating a larger financial cushion against extreme losses.

This matters because the Pareto claim-size distribution is heavy-tailed; most claims are manageable, but a few very large claims can cause severe losses. The profit-taking policy leaves the company more exposed to those events, creating a higher-risk “survive or fail” outcome structure. The no-profit-taking approach is therefore more stable in this simulation because it allows the company to retain capital and improve long-term solvency.